

Symbols, Constants and Units

EART23001 Atmospheric Physics and Weather

Using this reference. These are the symbols used most often in the practice sheets. You are not expected to memorise every numerical constant. Values needed for a calculation are normally supplied in the question. Units shown here are the preferred SI units; hPa, km, μm and other convenient atmospheric units are used where appropriate.

State, thermodynamics and vertical structure

Symbol	Meaning	Typical unit/value
p	pressure	Pa; 1 hPa = 100 Pa
T	absolute temperature	K
T_d	dew-point temperature	K or $^{\circ}\text{C}$
ρ	air density	kg m^{-3}
z	height above a reference level	m or km
g	acceleration due to gravity	9.81 m s^{-2}
R^*	universal gas constant	$8.314 \text{ J mol}^{-1} \text{ K}^{-1}$
R_d	specific gas constant for dry air	$287 \text{ J kg}^{-1} \text{ K}^{-1}$
R_v	specific gas constant for water vapour	$461.5 \text{ J kg}^{-1} \text{ K}^{-1}$
c_p	specific heat capacity of dry air at constant pressure	$1004 \text{ J kg}^{-1} \text{ K}^{-1}$
H	pressure/density scale height in a simplified exponential atmosphere	m or km
Γ_{env}	environmental lapse rate, $-dT/dz$	K km^{-1}
Γ_d	dry adiabatic lapse rate	$\approx 9.8 \text{ K km}^{-1}$
Γ_s	saturated (moist) adiabatic lapse rate; not a universal constant	K km^{-1}
θ	dry potential temperature	K
κ	R_d/c_p	≈ 0.286

Water vapour and cloud microphysics

Symbol	Meaning	Typical unit/value
e	water-vapour partial pressure	Pa or hPa
e_s	saturation vapour pressure; subscripts w and i denote liquid water and ice	Pa or hPa
RH	relative humidity	%
r_v	water-vapour mixing ratio (mass of vapour / mass of dry air)	kg kg^{-1} or g kg^{-1}
q	specific humidity (mass of vapour / mass of moist air)	kg kg^{-1}
ϵ	ratio R_d/R_v	≈ 0.622
L_v	latent heat of vaporisation	$\approx 2.5 \times 10^6 \text{ J kg}^{-1}$
N	cloud-droplet number concentration	m^{-3} or cm^{-3}
r	droplet/hydrometeor radius	m or μm
ρ_w	density of liquid water	1000 kg m^{-3}
LWC	liquid-water content (liquid-water mass per volume of air)	kg m^{-3} or g m^{-3}
M_2	area-related second moment. For identical droplets, $M_2 = Nr^2$	m^{-1} if N is m^{-3}
M_3	volume/mass-related third moment. For identical droplets, $M_3 = Nr^3$	dimensionless in this simplified form
β_{ext}	volume extinction coefficient; for cloud droplets it is proportional to an area measure such as M_2	m^{-1}

τ	optical depth, approximately $\beta_{ext}\Delta z$ for a uniform layer	dimensionless
A_c	cloud albedo (fraction of incident shortwave radiation reflected)	dimensionless
σ_w	surface tension of liquid water	$\approx 0.075 \text{ N m}^{-1}$ near 10°C
η	dynamic viscosity of air	$\approx 1.7 \times 10^{-5} \text{ Pa s}$
v_t	terminal fall speed	m s^{-1}
E	collection efficiency used in collision-coalescence calculations	dimensionless, $0 \leq E \leq 1$

Atmospheric dynamics

Symbol	Meaning	Typical unit/value
u, v, w	eastward, northward and vertical wind components	m s^{-1}
V	wind-speed magnitude when a component is not required	m s^{-1}
a_p	horizontal pressure-gradient acceleration	m s^{-2}
Ω	Earth's angular rotation rate	$7.292 \times 10^{-5} \text{ s}^{-1}$
ϕ	latitude	degrees or radians
f	Coriolis parameter, $2\Omega \sin \phi$	s^{-1}
v_g	geostrophic wind speed (or a component of the geostrophic wind)	m s^{-1}
$\Delta p/\Delta x$	finite-difference horizontal pressure gradient	Pa m^{-1}

Radiation and planetary energy balance

Symbol	Meaning	Typical unit/value
$L_\odot$	solar luminosity	$\approx 3.83 \times 10^{26} \text{ W}$
d	distance from the Sun in inverse-square calculations	m
S_0	solar irradiance (solar constant) at mean Earth–Sun distance	$\approx 1361 \text{ W m}^{-2}$
F_{abs}	absorbed radiative flux	W m^{-2}
α	planetary/surface albedo unless a cloud-specific A_c is used	dimensionless
σ_{SB}	Stefan–Boltzmann constant	$5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
T_e	effective radiating temperature	K
T_s	surface temperature in an energy-balance model	K
T_a	atmospheric-layer temperature in a simple greenhouse model	K
λ	wavelength	$\text{m}, \mu\text{m} \text{ or } \text{nm}$
λ_{max}	wavelength of maximum blackbody emission	$\text{m} \text{ or } \mu\text{m}$
b	Wien displacement constant	$2.898 \times 10^{-3} \text{ m K}$

Notation note. r_v is reserved for water-vapour mixing ratio, while r is used for particle radius. σ_{SB} denotes the Stefan–Boltzmann constant and σ_w denotes water surface tension. This avoids two common notation clashes in atmospheric physics.